

Ion implantation assisted synthesis of graphene on various dielectric substrates

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ABSTRACT

Direct synthesis of high-quality graphene on dielectric substrates is of great importance for the application of graphene-based electronics and optoelectronics. However, high-quality and uniform graphene film growth on dielectric substrates has proven challenging due to limited catalytic ability of dielectric substrates. Here, by employing a Cu ion implantation assisted method, high-quality and uniform graphene can be directly formed on various dielectric substrates including SiO₂/Si, quartz glass, and sapphire substrates. The growth rate of graphene on the dielectric substrates was significantly improved due to the catalysis of Cu. Moreover, during the graphene growth process, the Cu atoms gradually evaporated away without involving any metal contamination. Furthermore, an interesting growth behavior of graphene on sapphire substrate was observed, and the results show the graphene domains growth tends to grow along the sapphire flat terraces. The ion implantation assisted approach could open up a new pathway for the direct synthesis of graphene and promote the potential application of graphene in electronics.

KEYWORDS

graphene, chemical vapor deposition (CVD), ion implantation, dielectric substrates

1 Introduction

Graphene, a two-dimensional atomic-crystallite material, possesses excellent physical properties [1, 2], making it attractive for use in nanoelectronics [3], optoelectronic [4, 5], and transparent electrodes [6]. Controllable growth of high-quality graphene is a significant challenge to fully realize the potential applications. Among various graphene synthesis approaches, the chemical vapor deposition (CVD) on metallic substrates has shown far greater success in the scalable production of uniform graphene [7–9]. However, in most applications, the transfer of graphene off the metals onto dielectric substrates unavoidably introduces defects, wrinkles and contaminations, which largely limits the performance of graphene-based electronic devices [10–12]. Therefore, to solve these issues, the way toward developing transfer-free routes for the direct growth of graphene on dielectric substrates is in great demand for various device applications.

To date, there are two approaches to prepare graphene on various dielectric substrates such as SiO₂ [11, 13], quartz glass [14–16], Si₃N₄ [17], Al₂O₃ [18–20], SrTiO₃ [21], MgO [22], and h-BN [23]. The first approach is metal-assisted growth, in which an upstream metal catalyst is used to decompose the carbon feedstock, enabling graphene to form on the downstream dielectric substrates [24, 25]. Alternatively, a thin metallic film or liquid metal film on dielectric substrates evaporate away at high temperature promoting the formation of graphene [26]. However, common metal-assisted approaches involved metal residue issues, which is fatal to semiconductor devices [12]. The second approach is to fabricate graphene directly or use plasma-enhanced method. The direct growth and plasma-

enhanced approaches show great advantages in terms of no metal contamination, but graphene on dielectric substrates still suffers from many issues of low quality, slow growth rate and uncontrollable thickness owing to the poor catalytic activity [15]. Therefore, the common metal-assisted technique or direct growth of graphene on dielectric substrates, albeit extremely challenging, is of great scientific and practical significance.

Ion implantation, a mature technology in the microelectronic industry, has been adopted by material field in semiconductor doping and surface modification [27]. In recent studies, ion implantation has been successfully applied to the preparation, doping and defect control of graphene [27–29]. Here, a novel approach using Cu ion implantation assisted atmospheric pressure CVD (APCVD) synthesis of high-quality graphene on various dielectric substrates such as SiO₂/Si, quartz glass, and sapphire has been demonstrated. In this approach, the Cu atoms enhanced the decomposition efficiency of hydrocarbon and completely evaporated away during the growth process of graphene, suggesting that there were no issues with contamination or doping of graphene. On quartz glass, significant changes have taken place assisted by Cu ion implantation: the graphene growth rate was enhanced to 6 times, while growth rate of the graphene domain size increased by 2.5 times. In addition, a new growth phenomenon that graphene domains grew along the flat terraces of sapphire substrates was observed for the first time. Given that ion implantation has been widely used in semiconductor technology, the approach of graphene growth on dielectric substrates assisted by Cu ion implantation will promote the development of graphene in the semiconductor industry.

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2 Experimental section

2.1 Materials

The dielectric substrates (SiO_2/Si , quartz glass, sapphire) were purchased from Shanghai Daheng Optics and Fine Mechanics Co., Ltd. The thickness of SiO_2 layer on Si wafer is about 285 nm, and the orientation of sapphire is C-plane (0001). The surface roughness of all dielectric substrates is below 10 Å.

2.2 Growth of graphene

Firstly, dielectric substrates (SiO_2/Si , quartz glass, sapphire) were thoroughly cleaned with deionized water, acetone, and ethanol in sequence. The Cu ion implantation was conducted at room temperature (RT) with the ion energy of 5 keV and the fluence of 2×10^{16} ions cm^{-2} on various dielectric substrates by employing a home-built ion implanter (BNU-400) at Beijing Radiation Center. Then, the implanted substrates were loaded into a commercially available tube furnace (Xiamen G-CVD Material Technology Co., Ltd.) with a quartz tube (2 in. diameter). In a typical APCVD process, the graphene growth was carried out at 1,100 °C with 200 sccm Ar, 50 sccm H_2 , and 3–5 sccm CH_4 for 60–300 min. The graphene layer can be tailored by varying the flow rate of CH_4 .

2.3 Transfer of graphene

Typically, the graphene film was transferred onto the holey carbon TEM grids by a standard wet transfer method using an etching solution (Hydrofluoric acid/DI water/ethanol with concentration 1:2:1) to dissolve the quartz glass substrates.

2.4 Characterization of graphene

The prepared samples were systematically characterized by optical microscopy (Olympus BX51), Raman spectroscopy (Thermo DXRxi, with laser excitation at 532 nm, and power of 5 mW), SEM (Hitachi S-4800, 1.0 kV), XPS (AXIS Supra/Ultra, using a monochromatic Al K α X-ray source), TEM (FET Tecnai F20, operating at 200 kV), and AFM (Asylum Cypher VRS, under tapping mode in ambient conditions). Transmittance was measured by UV–vis spectroscopy (PekinElmer Lambda 950 spectrophotometer). The sheet resistance of graphene films was tested by a four-probe resistance measuring meter (CDE, ResMap 178) based on the four-point probe method to eliminate contact resistance. Four metal probes were aligned in a line at intervals of 1 mm. For the temperature measurements of graphene-based heaters, a infrared thermometer (HCJYET, HT-8830) was employed.

3 Results and discussion

Figure 1(a) shows a schematic diagram of the Cu ion implantation assisted APCVD approach. The Cu ions were implanted into the dielectric substrates (SiO_2/Si , quartz glass, sapphire) with the energy of 5 keV, and then graphene was synthesized on the dielectric substrates by APCVD at 1,100 °C for 60–300 min. In high temperature condition, the Cu atoms gradually diffused to the substrate surface and effectively promote the nucleation and growth of graphene on the dielectric substrates. As the growth time increases, the Cu atoms evaporated gradually during the graphene growth. The inset images in Fig. 1(b) display that uniform monolayer graphene films were obtained on the different dielectric substrates (SiO_2/Si , quartz glass, sapphire) by using Cu ion assisted APCVD approach. Furthermore, the monolayer feature of as-grown graphene on different dielectric substrates was confirmed by Raman spectra [30]. It is known that the ratio of the 2D peak to G peak (I_{2D}/I_G) reflects the graphene layer thickness. As shown in Fig. 1(b), the I_{2D}/I_G values of as-grown graphene on SiO_2/Si , quartz glass, and sapphire substrate were 1.68, 1.71, and 1.41, respectively, indicating the formation of high-quality monolayer graphene on the various dielectric substrates.

Figure 2(a) displays representative optical images of the quartz glass substrates before (pristine glass) and after graphene growth with different graphene layers. These images show uniform color contrasts and controllable optical transparency. Figure 2(b) shows the typical Raman spectra of as-grown graphene glass by APCVD at 1,100 °C for 300 min, showing that the graphene layer can be effectively tailored by varying flow rate of CH_4 . Figure 2(c) displays the optical transparencies and sheet resistances of the as-grown graphene on quartz glass with different layers, revealing the ability to tailor the optical and electrical properties of graphene glass. The optical transparency was tested by UV–vis transmittance spectroscopy measurement, and the sheet resistance was measured by the four-probe method without any pretreatment of as-grown graphene glass. At a wavelength of 550 nm, the transmittances of as-grown monolayer, bilayer, and multilayer graphene on quartz glass were 95.2%, 89.5%, and 80.3%, along with sheet resistance of 7.84, 4.10, and 2.31 $\text{k}\Omega/\text{sq}$, respectively. The sheet resistance mapping of 2 cm $\times$ 2 cm multilayer graphene glass (transparency of 80.3% at 550 nm) displays uniform sky blue in most of the areas, revealing the excellent uniformity of as-grown graphene, as displayed in Fig. 2(d). Interestingly, the as-grown graphene on quartz glass would dramatically tailor the surface hydrophobicity of quartz glass, which can be characterized

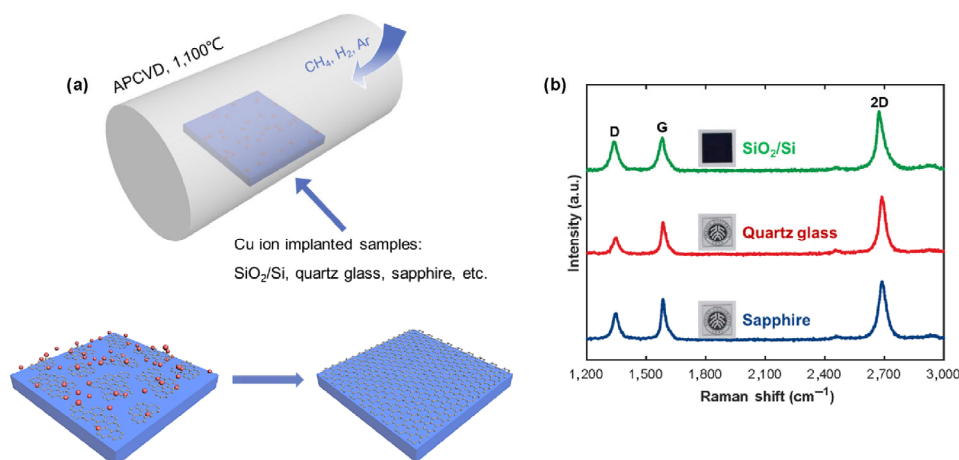


Figure 1 Ion beam assisted synthesis of graphene on SiO_2/Si , quartz glass and sapphire substrates. (a) Schematic diagram of the Cu ion assisted APCVD process. (b) Photograph and Raman spectra of as-grown graphene on SiO_2/Si , quartz glass, and sapphire substrates by APCVD at 1,100 °C for 300 min.

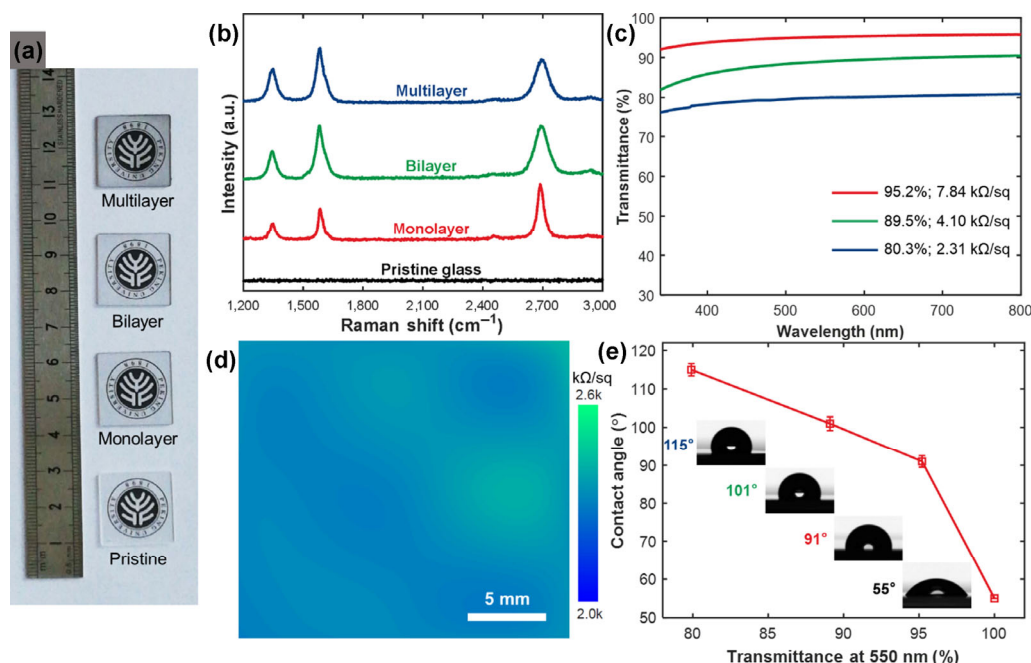


Figure 2 (a) Photograph and (b) Raman spectra of the quartz glass substrates before (pristine glass) and after graphene growth by APCVD at 1,100 °C for 300 min with different graphene layers by varying the flow rate of CH₄. (c) Correlation of UV–vis transmittance and sheet resistance of the graphene on quartz glass corresponding to the previous Raman spectra. (d) Mapping of the sheet resistance on as-grown graphene glass with a transparency of 80.3%. (e) Contact angle versus optical transmittance of pristine quartz glass and as-grown graphene glass samples.

by the static contact angle of pure water droplets. Figure 2(e) shows the hydrophobic characteristics of as-grown graphene glass, as indicated by the plot of the contact angle value versus optical transmittance. The contact angle remarkably enlarges with increasing the layer of graphene, reaching 115° with a transmittance (at 550 nm) of 80.3%. The difference in the surface wetting behaviors between the pristine and graphene glass was prominent, showing contact angle value changed from 55° to 115°. Therefore, the hydrophobic feature of graphene glass is essential for practical applications such as self-cleaning windows and anti-icing glass [14, 16].

To further investigate the graphene layer of the as-grown monolayer, bilayer, and multilayer graphene on quartz glass, the transferred graphene films on holey carbon TEM grids were

characterized by transmission electron microscope (TEM). Figures 3(a1)–3(c1) show typical low-magnification micrograph of graphene films. Figures 3(a2)–3(c2) show the high-resolution TEM images obtained from the crack edges of the graphene in the red box of Figs. 3(a1)–3(c1), respectively. The clear one edge line in Fig. 3(a2) and two lines in Fig. 3(b2) indicate the formation of monolayer and bilayer graphene. Similarly, the multilayer graphene can be identified from the intensity profiles in the inset of Fig. 3(c2). It is also worth noting that as the layer number of graphene increased, the high CH₄ flow was required to prepare multilayer graphene. However, the limited catalytic ability of Cu atoms would lead to severe amorphous carbon contamination on the surface of the multilayer graphene, which was also evidenced by Raman D peak in Fig. 2(b). It

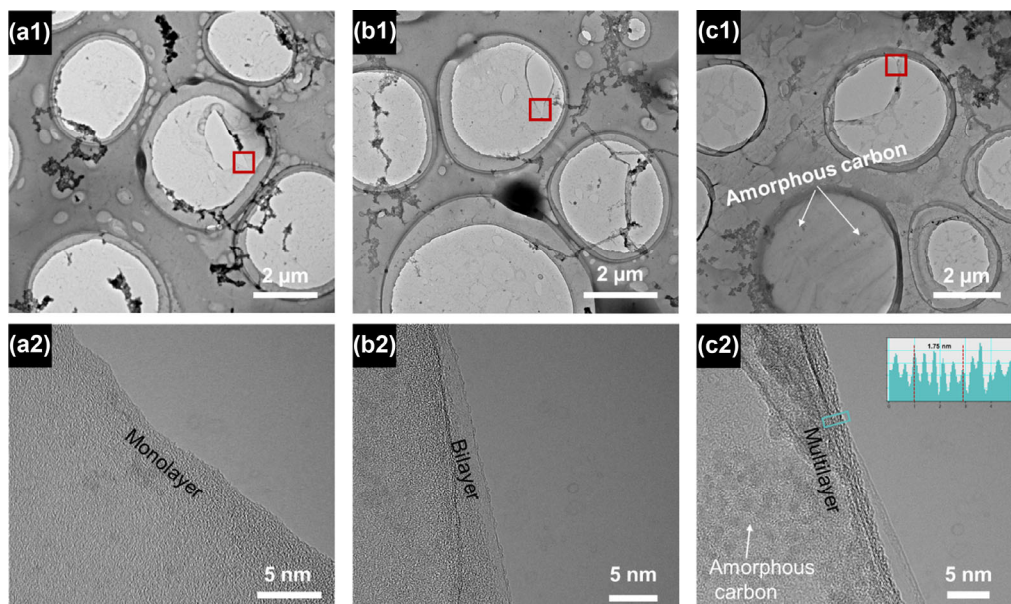


Figure 3 (a1)–(c1) Plan-view TEM images of monolayer, bilayer and multilayer graphene transferred from quartz glass, respectively. The high-resolution TEM images (a2)–(c2) were obtained from the crack edges of the graphene in the red box of (a1)–(c1), respectively.

should be noted that the layer number of graphene growth on dielectric substrates was largely dependent on the flow of CH_4 , and most areas of graphene were uniformly grown.

In order to study the role of the Cu atoms in graphene growth, X-ray photoelectron spectroscopy (XPS) was employed to measure the signal of Cu before and after the growth of graphene on the surface of Cu implanted quartz glass. As shown in Fig. 4(a), both the Cu 2p peak and Cu LMM region completely disappeared after the growth of graphene at high temperature and proved that the Cu atoms could gradually evaporate from the surface of the implanted substrates. Figure 4(b) shows that the Cu 2p spectrum was deconvoluted into two major components that were assigned to Cu 2p_{3/2} peaks in Cu (933.0 eV) and CuO (934.2 eV), Cu 2p_{1/2} peaks in Cu (952.3 eV) and CuO (954.1 eV), and strong Cu²⁺ satellite peaks [31, 32]. Moreover, by taking the area ratio of the peaks Cu 2p_{3/2} peak in Cu (green region) and CuO (blue region), the implanted Cu atoms were partially oxidized and the ratio of Cu:CuO was about 5:4 in the Cu implanted samples. According to the results above, the CuO can be reduced to Cu atoms by hydrogen at high temperature and further promote the nucleation and growth of graphene on the dielectric substrates. The most critical issue is the amount of Cu atoms was very low and the Cu atoms completely disappeared after graphene growth, indicating that the Cu ion implantation assisted approach will not contaminate or dope graphene.

In order to investigate the Cu ion implantation assisted growth behavior of graphene, the evolution of the graphene domains on the dielectric substrates at different growth stages was characterized by SEM with and without Cu ion implantation. Figures 5(a)–5(c) clearly display the different growth stages of graphene on quartz glass assisted by Cu ion implantation for 60, 120, and 180 min, respectively. With the aid of Cu atoms, a lot of graphene domains began to nucleate within 60 min (Fig. 5(a)), and graphene domains grew rapidly until covered the entire surface (Figs. 5(b) and 5(c)) on the Cu implanted quartz glass. In contrast, the nucleation time and the growth rate on untreated quartz glass were significantly lower than the Cu implanted quartz glass (Fig. S2 in the Electronic Supplementary Material (ESM)). In the untreated quartz glass sample, no graphene domains were observed until the growth time increased to 180 min (Fig. S2(a) in the ESM) and it took 480 min or more to cover the entire surface (Fig. S2(c) in the ESM). The rate of average graphene coverage and the average

graphene domain area with Cu ion implantation were about 6 and 2.5 times that of without Cu ion implantation, respectively (Figs. 5(d)–5(f)). This proved that the Cu ion implantation could effectively reduce the nucleation time and promote the growth rate of the graphene domains. In addition, there was a very interesting graphene growth phenomenon observed on the sapphire substrates. At the early stage of graphene growth on sapphire substrates, graphene domains seemed to be distributed randomly across the entire surface (Fig. 5(d) and Fig. S4 in the ESM). However, with the increase of growth time, all the graphene domains growth tended to grow along the flat terraces pointed by the red dashed lines (Figs. 5(e) and 5(f)). It is considered that the reason causing this difference from quartz glass is the most commercially available sapphire crystal substrates polished at an off-axis (or “tilt-angle”) angle off the (0001) basal plane, resulting in a surface with atomic steps and flat terraces between the steps as schematic diagram in Fig. S3 in the ESM [33]. The round graphene domains on flat terraces were formed because isodirectional (disk-like) growth on flat terraces of nonmetal substrates has the lowest energy atomic configuration at nucleating graphene edges [13]. Interestingly, the average diameter of the graphene domains (260 ± 12 nm) is close to the width of the flat terraces (280 ± 10 nm), indicating the graphene domain size depends on the flat terraces of sapphire substrates. This is the first time that graphene domains along the flat terraces of sapphire substrates have been seen during the graphene growth process, which is of great significance for improving the quality of graphene on dielectric substrates and applications in optoelectronic devices and LED illumination [33].

To explore practical applications of the conductive graphene on dielectric substrates, the as-grown graphene samples were tested in a series of heating thermochromic display devices. A piece of 2 cm × 2 cm as-grown graphene glass was processed with deposition of conductive silver paste, a color reversible thermochromic ink (red to white at 31 °C), and carbon tape electrodes. Figures 6(a1)–6(c1) show schematic diagrams of the operation procedure of the graphene/quartz glass-based thermochromic display. When an input voltage of 30 V was applied, the red color changed to white within 21 s (Figs. 6(a2)–6(c2)), then the color returned to red when the input voltage was removed (Fig. 6(c2)). The surface temperature mapping of the whole device between 40 and 45 °C at an ambient temperature of 25 °C indicates the uniformity of graphene

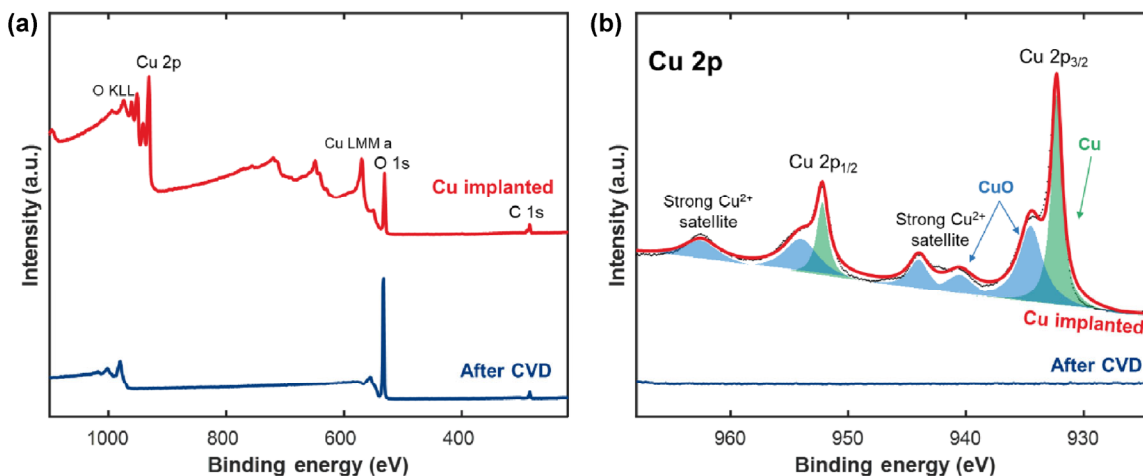


Figure 4 (a) XPS survey spectra and (b) Cu 2p spectra before and after the growth of graphene on the surface of Cu implanted quartz glass. The Cu 2p peak can be deconvoluted into two major components, corresponding to Cu and Cu²⁺ states along with strong Cu²⁺ satellite peaks, which were filled with green and blue, respectively.

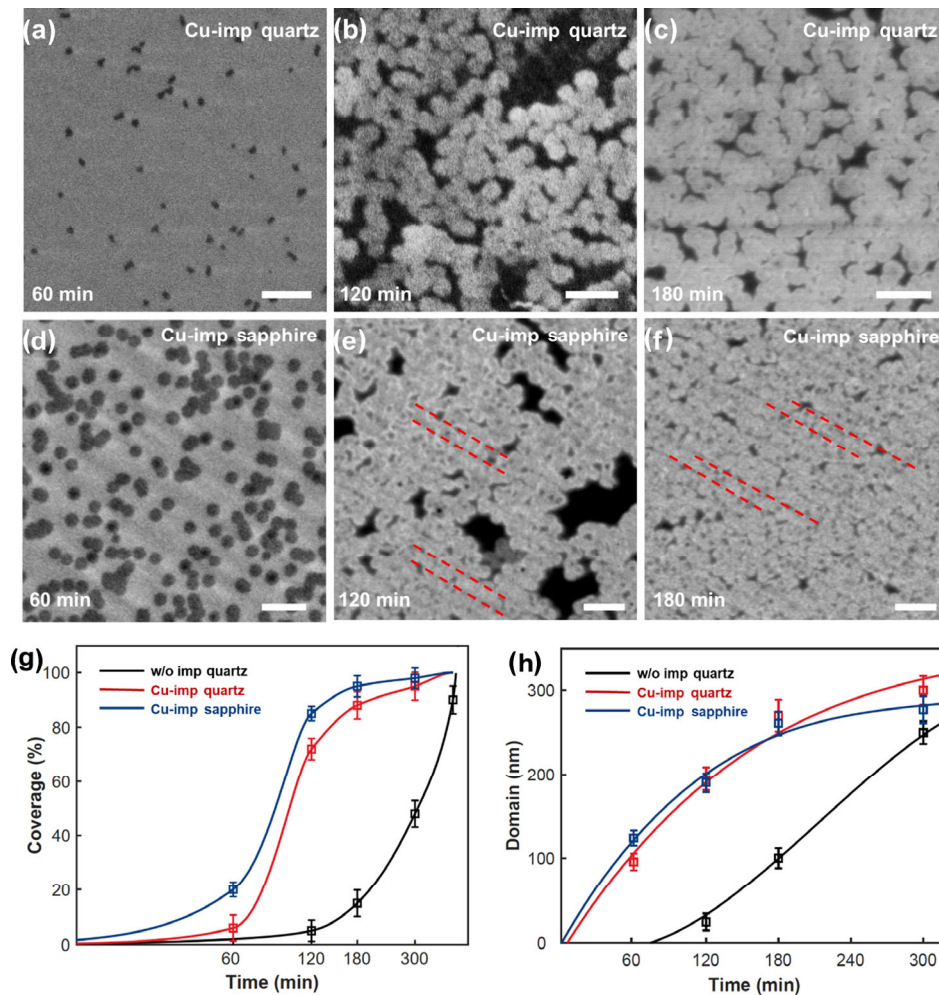


Figure 5 (a)–(c) SEM images of graphene film growth assisted by Cu implantation at different stages for 60, 120, and 180 min on quartz glass and (d)–(f) sapphire substrates, respectively. The red dashed lines in (e) and (f) show graphene grow along the flat terraces of sapphire crystal. All scale bars in SEM images are 500 nm. (g) Coverage of graphene versus growth time on pristine quartz glass, Cu implanted quartz glass and Cu implanted sapphire substrates. (h) Time evolution of graphene domain size on Cu implanted quartz glass and Cu implanted sapphire substrates.

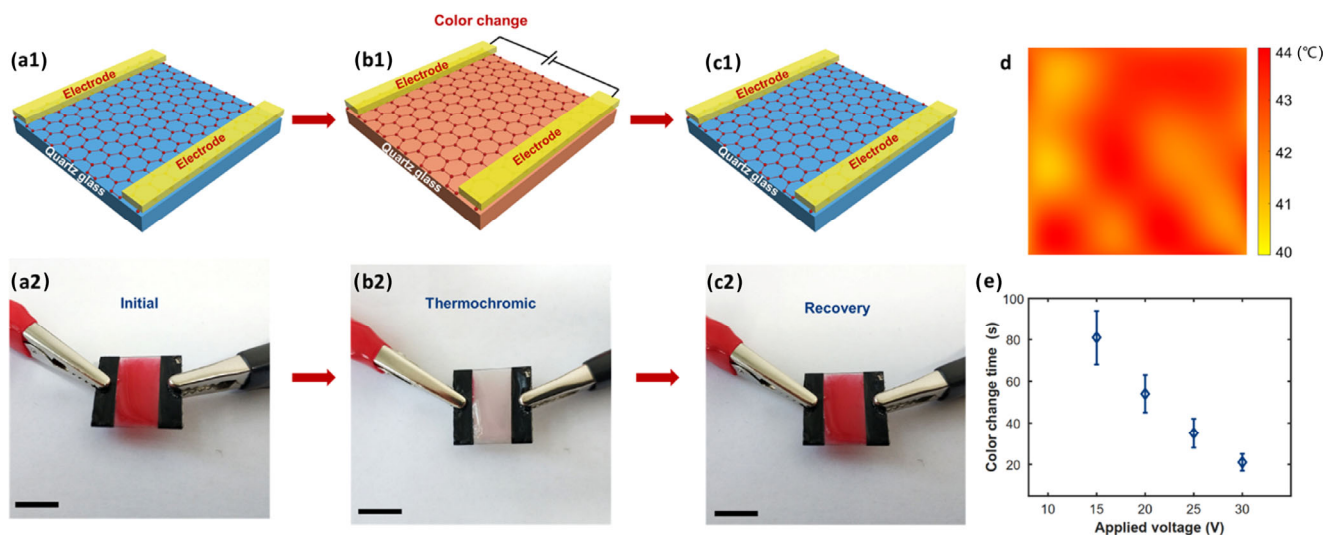


Figure 6 Application of as-grown graphene on quartz glass as a heating thermochromic display device. (a1)–(c1) Schematic diagrams showing the operation procedure of the graphene/quartz glass-based thermochromic display. (a2)–(c2) Photograph of the thermochromic display, showing its initial, thermochromic and recovery state at bias of 30 V. Conductive silver paste was deposited on the edges of the graphene/quartz glass. All scale bars in photograph are 1 cm. (d) Surface temperature mapping of graphene glass at an input voltage of 30 V. (e) Color change time as a function of input voltage.

heating device (Fig. 6(d)). For thermochromic display devices, the color change time decreases with increasing input voltage, achieving less than 40 s at 30 V (Fig. 6(e)), outperforming other

graphene-based thermochromic display devices. Furthermore, the cycling performance of the thermochromic display device on our as-grown graphene/quartz glass was tested, the color

change time remained the same after dozens of cycles.

4 Conclusion

In summary, an ion implantation assisted CVD technique was demonstrated to synthesize high-quality graphene on various dielectric substrates including SiO₂/Si, quartz glass, and sapphire substrates. Due to the catalysis of Cu, graphene film was directly formed on the dielectric substrates, in the meantime, the Cu atoms gradually evaporated away without involving any metal contamination. On the sapphire substrate, our results firstly reveal that graphene domains grow along the sapphire flat terraces during the growth process. In addition, the graphene grown on quartz glass was suitable for self-cleaning window or heating device in potential applications. Thus, the ion implantation technology provides an effective path in the synthesis of graphene on dielectronic substrates for further applications of electronics and optoelectronics.

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